

Chia-Wei Lin



CONTACT

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EDUCATION

National Yang Ming Chiao Tung University | 2025-
Electronic Engineering

- Professor : Jiun-In Guo
- Lab : iVsLab (Intelligent Vision System Design Lab)
- GPA : 4.05
- **2026 Summer Intern, CN5 Division, Realtek Semiconductor Corp.**



National Taipei University of Technology | 2021-2025
Electronic Engineering

- Rank: 4/109
- **President's Award (3-time recipient)**
- Bronze Medal, 2024 Creative Tech Challenge
- Honorable Mention, National Practical Project Competition
- 2024 Summer Intern, Zhen Ding Technology



RELATED COURSES

National Yang Ming Chiao Tung University
Integrated Circuit Design Lab (ICLAB) A+
VLSI Design and Implementation **A+**
Memory Circuit and System Design **A**
Digital integrated circuits **A**
Digital Signal Processing **A**
Computer Architecture **A-**

National Taipei University of Technology
Integrated Circuit Analysis and Simulation **A+**
FPGA System Design **A+**
Computer Architecture **A+**
Very Large Scale Integration **A**
Digital Signal Processing Laboratory **A+**

PROJECTS

Motion Vector Difference Matching (ICLAB Final Project)

- Introduction: Developed a Motion Estimation engine featuring 6-tap FIR half-pixel interpolation and SATD (Hadamard Transform) with mirror search.
- Optimization: Engineered 10-way parallel interpolation and a 4-stage pipeline. Reduced average latency from 200 to 39 cycles via early-calculation/output strategies.
- Takeaway: Mastered full-flow Cell-Based design (**RTL to APR**).
- Result : **Rank:5** / 1st demo / 99.01

H.264 Lite Prediction and Transform Engine

- Introduction: Implemented H.264 Intra 4x4/16x16 prediction, including integer transform, quantization, and a reconstruction loop for reference pixel updates.
- Optimization: Minimized latency by implementing the Hadamard matrix with add/sub logic and overlapping reconstruction calculations with I/O operations.

PPG-Based Wrist Monitoring Device

- Introduction : A wearable device utilizing PPG (infrared) and ML regression to monitor Blood Pressure, HR, SpO2, and HRV.
- Role: **Team Lead**. Developed ML regression models and MCU-to-PC communication

MIPS CPU 、 Convolution Neural Network 、 Multi-Packet Channel Arbiter

SKILL

Programming Language
Verilog/System Verilog
Python

EDA Tools
Synopsys : Verdi、VCS、PrimeTime
Cadence : Innovus、Jasper

Cell Based Design
CDC、Low Power、STA、APR、SYNTHESIS